MID-STATE TECHNICAL COLLEGE · IT PROGRAMS

Blocking Unknown Threats with Advanced WildFire

Unit 11 · Information Security 2 (10-151-111)

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Learning Outcomes

By the end of this unit, you will be able to:



Explain WildFire purpose and operation

Trace how the firewall and WildFire move a file from hash to verdict to signature.



Compare licenses and deployment choices

Distinguish basic, WildFire, and Advanced WildFire, and public, private, and hybrid clouds.



Configure and update WildFire

Set general settings, build and attach a WildFire Analysis Profile, and schedule updates.



View WildFire reports and logs

Verify uploads, read verdicts and analysis reports, and dispute an incorrect verdict.

LESSON 1

WildFire Concepts



Why Signatures Aren’t Enough

- A **known threat** matches an existing antivirus signature. The Antivirus profile from Unit 9 stops it.
- An **unknown threat** has no signature yet: a zero-day, or a fresh variant packed to look new.
- Modern malware is **customized for a specific network**, so it is often the first and only copy anyone has seen.
- A **sandbox** runs a suspicious file in an isolated environment and watches what it actually does.



The signature gap

Signature-only defenses are always one step behind. The gap between a new sample appearing and a signature existing is exactly where targeted attacks live.

The WildFire Threat Intelligence Cloud

Firewalls around the world feed WildFire, and every subscriber benefits from what it learns.

Samples flow in

- Firewalls automatically forward **unknown files** and **email links**
- Licensed users upload samples through the **WildFire portal** or **API**
- Samples run in Windows, macOS, Linux, Android, and bare-metal environments

Protection flows out

- Malware gets a new **antivirus signature**
- Phishing links are added to the **PAN-DB** phishing category
- Subscribed firewalls worldwide download the update within minutes

Network effect: malware first detected anywhere becomes protection everywhere, regardless of where the sample was first seen.

How WildFire Reaches a Verdict

The firewall runs quick checks first. Only truly unknown files within the size limit are forwarded.

Check	What happens
Trusted signer?	Files signed by a trusted signer are allowed and not forwarded
Hash the file	The firewall hashes the file and asks WildFire if it has seen it
Already seen?	The existing verdict is reused; the file is not resubmitted
Size limit	Files over the configured maximum are allowed but **not analyzed**
Analysis	WildFire runs the sample, generates a verdict, and informs the firewall
Protection	Malware signatures reach subscribers in minutes, everyone else in 24–48 hours

Signatures for unlicensed firewalls arrive with the daily Antivirus content update.

WildFire Verdicts

Every file or link WildFire analyzes gets one of four verdicts:

Verdict	Meaning	What WildFire does
Benign	Safe; no malicious behavior	Logged only if reporting is enabled
Grayware	Obtrusive but not a direct threat: adware, spyware, browser helper objects	No signature; can be logged
Malicious	Malware: viruses, worms, trojans, RATs, rootkits, botnets	Generates and distributes a signature
Phishing	Email link to a site that tries to steal credentials	Adds the URL to PAN-DB

Verdicts appear in Monitor > Logs > WildFire Submissions and in the WildFire portal.

WildFire Protects Email

- The firewall forwards **attachments** and the **HTTP/HTTPS links** found in SMTP and POP3 email.
- A malicious attachment produces a new **antivirus signature** pushed to firewalls.
- A malicious or phishing link is added to **PAN-DB** so URL Filtering can block it.
- **Email headers** appear in the WildFire Submissions log; with User-ID, they link to the targeted user in the ACC.



What is not stored

Neither the firewall nor WildFire stores or lets anyone view the email content. Only the attachment or link is analyzed.

Where WildFire Results Land

WildFire findings feed three different content databases, each tied to its own license:

Database	Contains	Timing	License
Antivirus	Signatures, including WildFire's	Daily; 24–48 hours	Threat Prevention
WildFire signatures	New WildFire malware signatures	Real time	WildFire
PAN-DB	Malware and phishing URLs	Looked up dynamically	URL Filtering

Common mistake: WildFire does not put antivirus signatures in PAN-DB. Signatures go to the antivirus databases; URLs go to PAN-DB.



LESSON 2

Licensing and Deployment

Basic, WildFire, and Advanced WildFire

	Basic WildFire	WildFire	Advanced WildFire
Cost	Included with firewall	Subscription	Subscription
File types	PE files only	Office, PDF, JAR, APK, ELF, archives, scripts, email links	Same as WildFire
Signature delivery	Antivirus update, 24–48 hrs	Real time	Real time
Inline ML on firewall	—	Yes	Yes
API, private and hybrid cloud	—	Yes	Yes
Evasive-malware engines	—	—	IRMA, unpacking, VMI, fingerprinting

Basic WildFire signatures require a Threat Prevention license to download.

Inside Advanced WildFire

Advanced WildFire layers extra detection on top of static, machine learning, and dynamic analysis:

Engine	What it defeats
Run-time Memory Analysis	Code that only reveals itself in memory while running
Automated Unpacking	Packing and obfuscation that hide the real payload
Stealthy Observation (VMI)	Malware that detects it is in an analysis VM
Dependency Emulation	Samples that refuse to run without expected DLLs
Malware Family Fingerprint	New variants of known families
Config File Analysis	Hidden command-and-control addresses (yields IOCs)

No extra firewall configuration: Advanced WildFire uses your existing WildFire Analysis Profile.

Intelligent Run-time Memory Analysis

Traditional approaches

- **Continuous capture** records everything but consumes heavy memory and processing
- **Intermittent capture** is cheaper but can miss the moment malware executes
- Both analyze after execution, from a fixed schedule

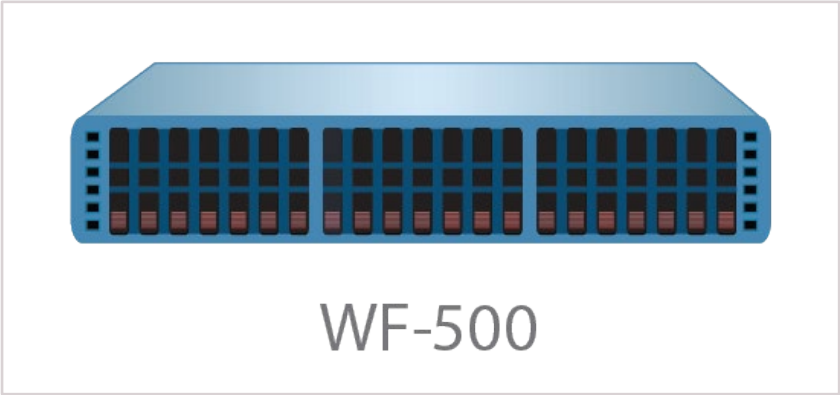
Intelligent capture (IRMA)

- Snapshots memory at **interesting events**, such as suspicious API calls
- Builds a **memory timeline** that exposes unpacked code
- Records permissions, OS structure changes, and API details as evidence

Why it matters for defenders: malware that never touches disk in readable form still leaves a trail in memory, and IRMA is built to catch it there.

Deployment Choices

- **Global public cloud** — the default, `wildfire.paloaltonetworks.com`, hosted in the U.S.
- **Regional public clouds** keep submissions in-region for privacy rules, e.g. `eu.wildfire.paloaltonetworks.com`.
- **Private cloud** — an on-premises **WF-500/WF-600** appliance; files never leave the network.
- **Hybrid cloud** — sensitive files go to the appliance, everything else to the public cloud.



WF-500 private cloud appliance

Hybrid Cloud: Who Analyzes What

Split files by sensitivity and by what the appliance supports:

File type	Destination	Reason
ms-office	Private	Likely confidential
pdf	Private	Likely confidential
pe, jar	Public	Rarely sensitive
apk	Public	Not supported on appliance
email-link	Public	Global link analysis



SECURITY FOCUS

Every file you forward to a public cloud leaves your network. Decide which file types may contain regulated or confidential data before you choose public-cloud for them.

Why it matters: Sending patient records or contracts to a third-party sandbox can itself be a compliance violation.

LESSON 3

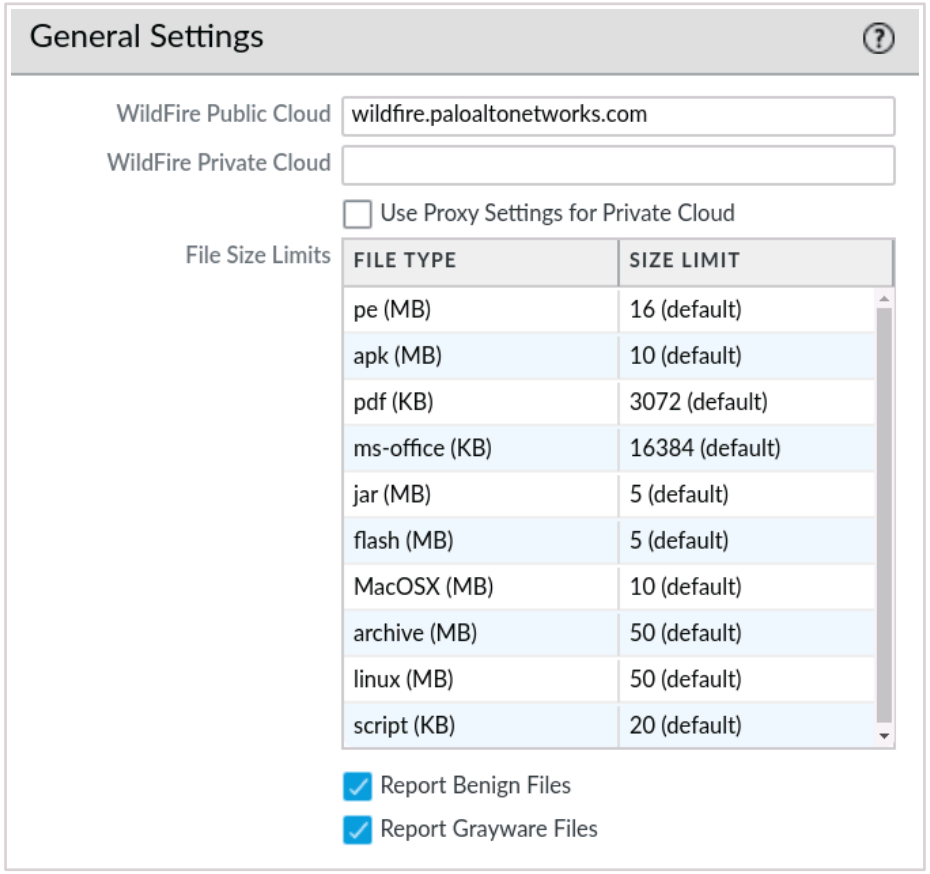
Configuring WildFire



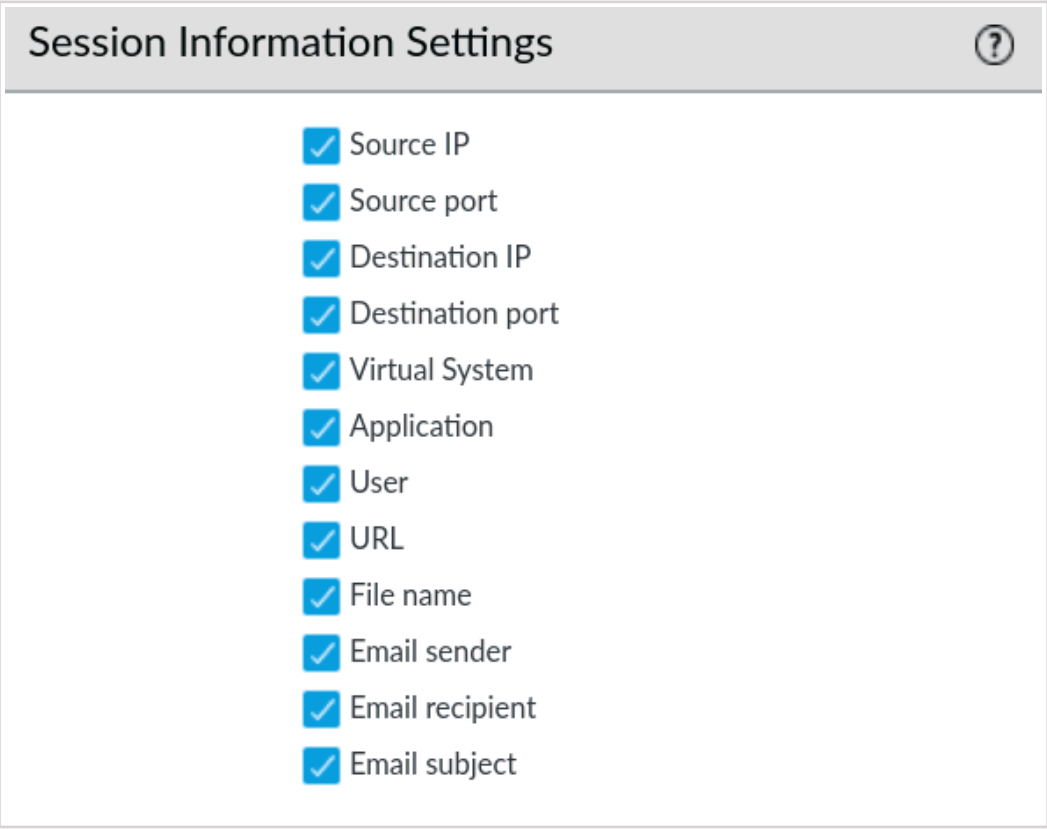
WildFire General Settings

Device > Setup > WildFire controls where files go and what comes back.

- **WildFire Public / Private Cloud** — the cloud URL or the appliance IP/FQDN
- **File size limits** per file type; larger files are never forwarded
- **Report Benign Files** and **Report Grayware Files** are off by default
- Decrypted traffic is not forwarded until you enable **Allow Forwarding of Decrypted Content** under Content-ID



Session Information Settings



- Chooses which **session details** are sent to WildFire with each sample.
- Everything is **selected by default**.
- What you send is what comes back in the **WildFire Submissions log** and portal report.

Trade-off

Clearing fields such as user or email recipient protects privacy, but removes the detail you need to find a victim.

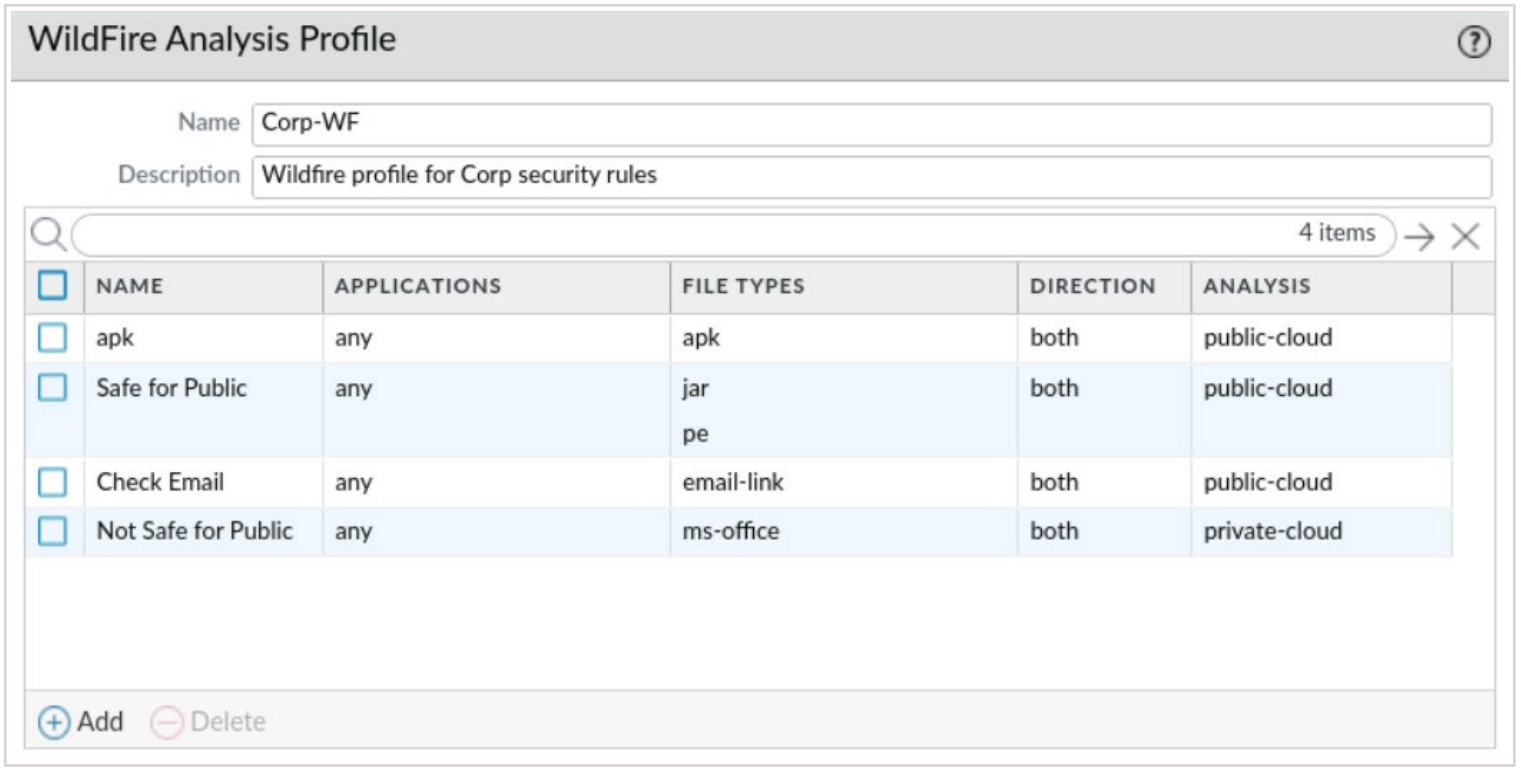
The WildFire Analysis Profile

Objects > Security Profiles > WildFire Analysis decides which files from allowed traffic are forwarded.

Field	Choices	Lab 10 value
Applications	Specific applications or any	`any`
File Types	pe, pdf, ms-office, apk, jar, email-link, … or any	`any`
Direction	upload, download, or both	`both`
Analysis	public-cloud or private-cloud	`public-cloud`

default profile: read-only, forwards all unknown files to the public cloud. Clone it or create a new profile to customize.

A Custom Profile Example



Rules send APK, JAR/PE, and email links to the public cloud, and Office files to a private appliance.

Attaching the Profile to Policy

- Add the profile to a **Security policy rule** directly, or to a **Security Profile Group** the rule uses.
- Only rules with an **allow** action need it; denied traffic is never inspected.
- The profile applies to every packet over the life of the session.
- Files sent are logged to **Monitor > Logs > WildFire Submissions**.



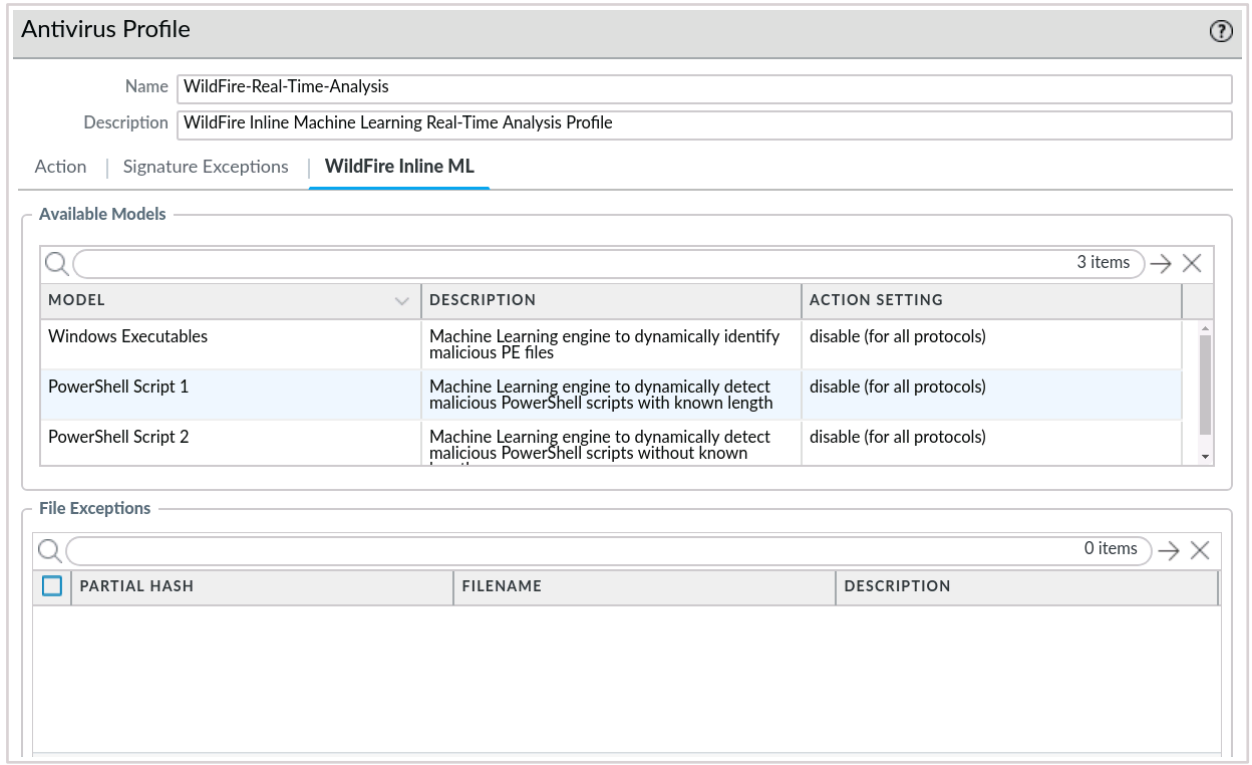
SECURITY FOCUS

If a File Blocking profile blocks a file type, that file is never forwarded to WildFire. A block action anywhere upstream means no verdict.

Why it matters: You can lose visibility into exactly the files attackers are trying to deliver.

WildFire Inline ML

Objects > Security Profiles > Antivirus — machine learning on the firewall, before a cloud verdict exists.

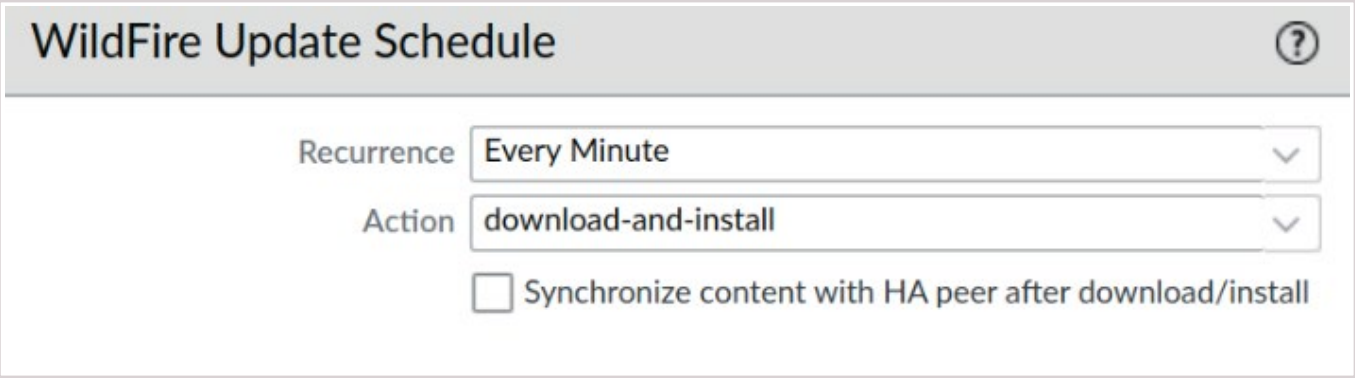


Action	Effect
enable	Use the per-protocol actions
alert-only	Log, but let traffic pass
disable	No inline ML action

Add a file exception by hash to exclude a known false positive.

WildFire Update Schedule

Device > Dynamic Updates controls how often the firewall checks for new WildFire signatures.



- **Real-time** updates are the recommended best practice with a WildFire license.
- Intervals from **every minute** up to once an hour are also available.
- Without a WildFire license, signatures still arrive in the **daily Antivirus** package.

Requires a license

The WildFire update schedule only appears with an active WildFire subscription.

LESSON 4

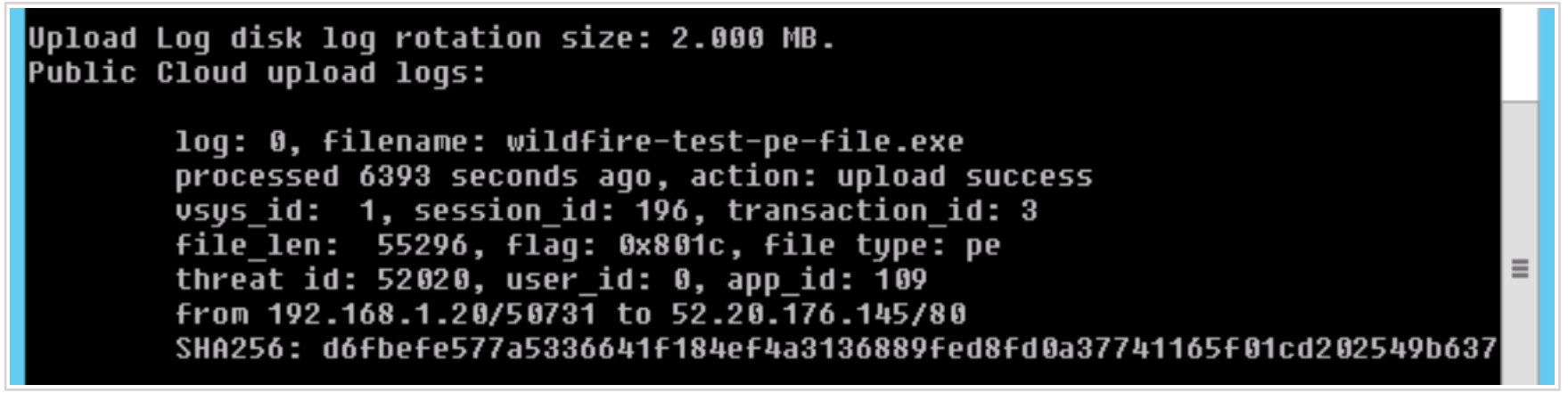
WildFire Reporting



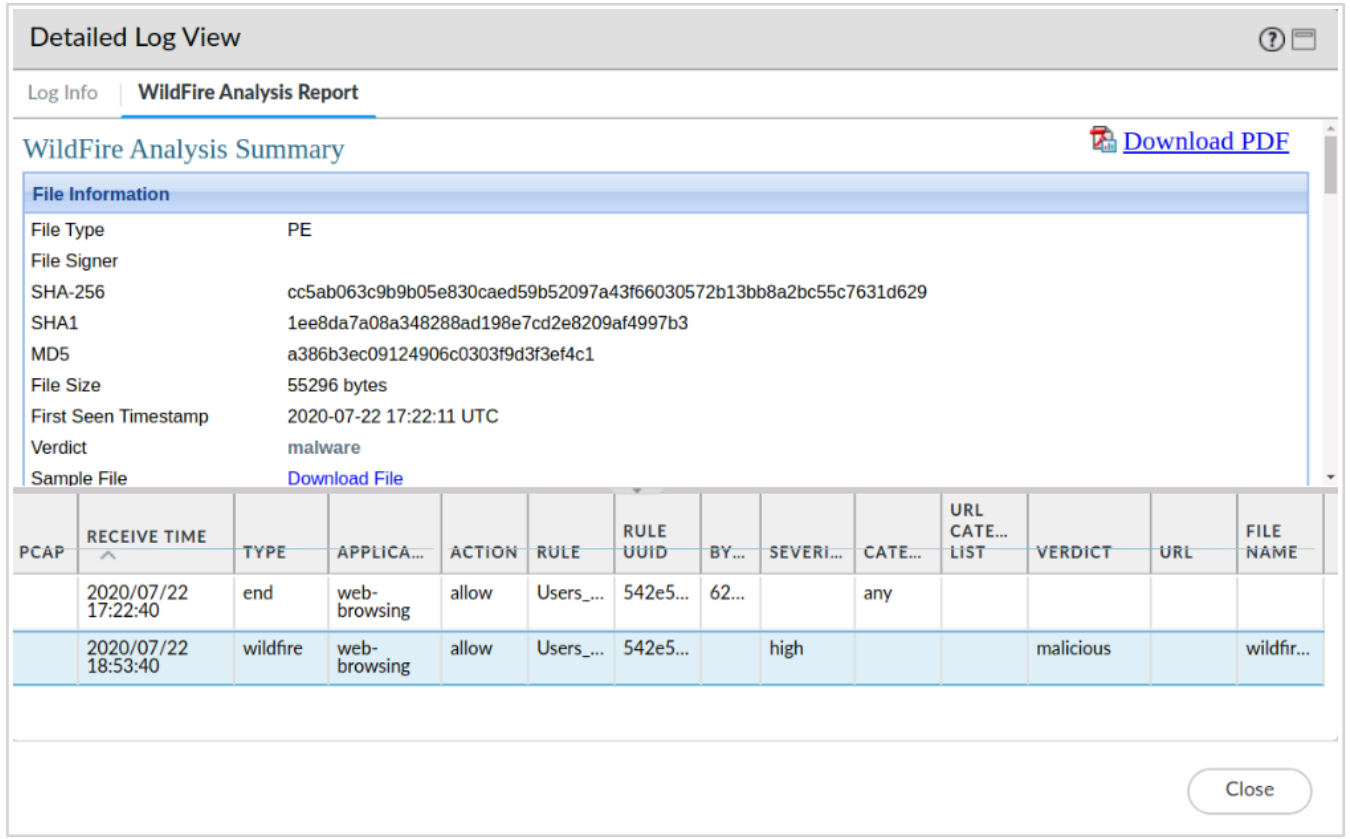
Verify the Upload from the CLI

Confirm the firewall actually sent the file to WildFire:

```
admin@firewall-a> debug wildfire upload-log show
```



The WildFire Submissions Log



- **Monitor > Logs > WildFire Submissions** lists one entry per analyzed sample.
- The **Verdict** column shows the result.
- Click the magnifying glass, then the **WildFire Analysis Report** tab.
- **Download PDF** for a full report with a behavior timeline.

Reading the Analysis Report

paloalto NETWORKS

WILDFIRE ANALYSIS REPORT

Download as PDF

FILE INFORMATION

File Type	JAVA Class
File Signer	
SHA-256	44c962009267ca4006c82b73ebed6478c8b9a6a4d264436904c112f13af222d6
SHA-1	e44fabc80be764c65f71a4ddd153c569a4f23003
MD5	bd3932e014535efcf56b67a1a113e2ed
File Size	2682 bytes
First Seen Timestamp	2022-11-21 10:57:52 PST
Sample File	Download File
Verdict	Malware

Questions it answers

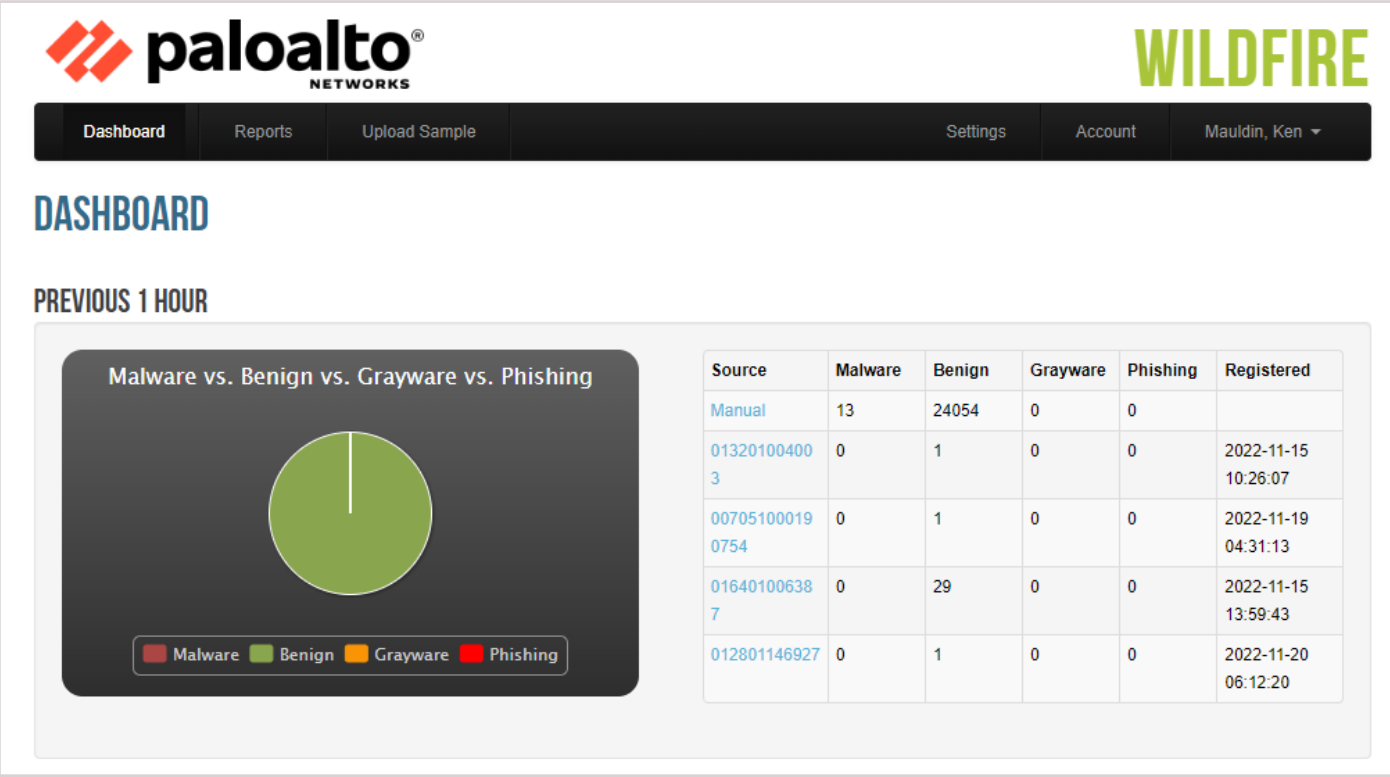
Who was **targeted**?

Which **application** carried the file?

What **behavior** did the sample show?

The PDF includes the analysis environment and a timeline of actions the malware took.

The WildFire Portal



- Browse to `wildfire.paloaltonetworks.com` with a support account.
- **Dashboard** summarizes verdicts for every firewall; click a serial number to filter.
- **Reports** searches all submissions; **Upload Sample** submits files manually.
- Set up **email notifications** under Settings.

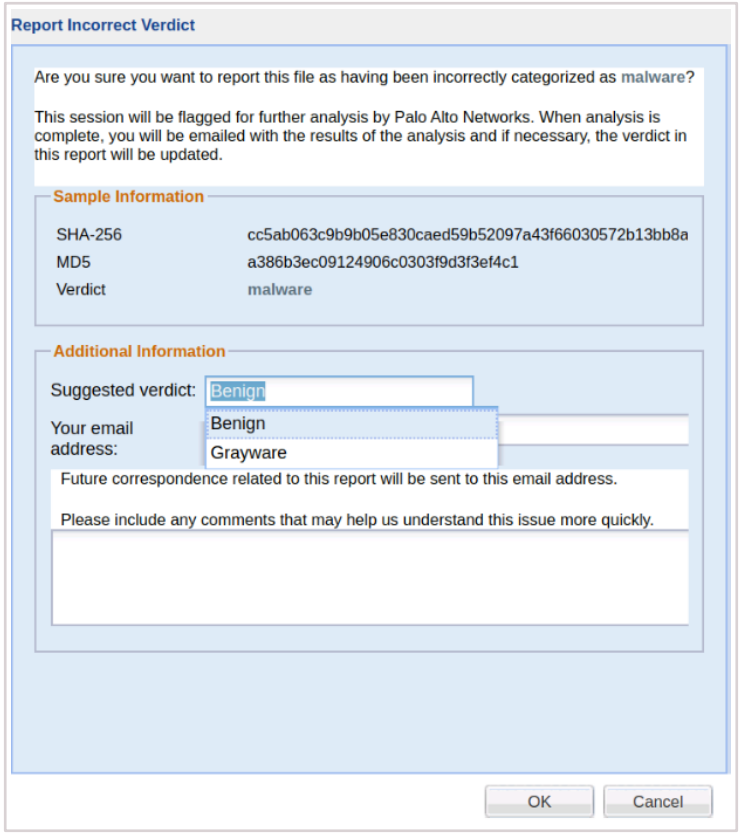
Report an Incorrect Verdict

From the firewall

- Monitor > Logs > WildFire Submissions
- Open the entry's detailed view
- Click the **WildFire Analysis Report** tab
- Use the incorrect-verdict link and suggest a verdict

From the portal

- Open the report from the Reports page
- Scroll to **Report an incorrect verdict**
- Fill in the suggested verdict and email
- Palo Alto Networks reviews the request



Security Focus: Where Files Slip Past WildFire

SECURITY FOCUS

Encrypted traffic — without decryption and Allow Forwarding of Decrypted Content, WildFire never sees the file.

File size limits — oversized files are allowed without analysis.

File Blocking upstream — blocked file types are never forwarded.

No profile on the rule — an allow rule without a WildFire profile forwards nothing.

Why it matters: Each gap is a silent one: traffic flows, nothing is logged, and you never learn what you missed.

Hands-On Lab 10: Blocking Unknown Threats

NETLAB+ · PAN-OS 11.0 Essentials · baseline edu-210-11.0a-10.xml



- Create the **Corp-WF** WildFire Analysis Profile with an **All_Files** rule to the public cloud.
- Add Corp-WF to **Corp-Profiles-Group** and set the other profiles to None.
- Enable Report Benign Files and Report Grayware Files, then commit.
- Download the test file and confirm the upload from the CLI.
- Examine the WildFire Submissions entry and the PDF analysis report.

Lab 10 Tips

Tip	Why
Use Firefox and http://	The test file URL is plain HTTP; match the lab guide exactly
Be patient	Analysis can take 5 to 15 minutes; keep refreshing the log
Don’t skip the rule edit	Set Allow-PANW-Apps source to Any — only in Detailed steps 2.4
Lab-only setting	Other profiles are off to isolate WildFire; keep them all on in production

Heads up: the High-Level steps number differently from the Detailed steps because Commit is its own step (1.5).

Check Your Understanding

- Which file type can a firewall send to WildFire without a WildFire subscription: PDF, JAR, APK, or EXE?
- Which verdict indicates obtrusive behavior but not a security threat?
- True or false: when WildFire finds a malicious email link, it updates antivirus signatures in PAN-DB.
- With a WildFire subscription, which file conditions could keep a file from being analyzed?



Unit 11 Homework

Always check Blackboard for assignments and due dates.



- **Unit 11 Lab** — NETLAB+ Lab 10: Blocking Unknown Threats with WildFire
- **Unit 11 Assignment** — ML-Based Prevention Video
- **Unit 11 Quiz**

Key Takeaways

- WildFire sandboxes unknown files and links, then turns malware into signatures shared worldwide.
- Basic WildFire covers PE files only; subscriptions add file types, real-time updates, and evasive-malware engines.
- Set general settings, attach a WildFire Analysis Profile to allow rules, and use real-time updates.
- Verify with the CLI, read the WildFire Submissions log, and dispute verdicts that look wrong.
- Decryption, size limits, and File Blocking all decide what WildFire never gets to see.

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Questions?

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